Qn 4	Answer	Marks
(a)(i)	Power supplied by battery = Energy / time = (11.3 J) /(10 x 60 s) = 0.0188 W	**M1** **A1**
(a)(ii)	From graph, resistance of thermistor = 3.1 kΩ Power dissipated through the resistor and thermistor = Power supplied by battery $\Rightarrow$ 0.0188 = E^2 /(3100+1200) E = 8.99 V	**B1** **M1** **A1**
(b)	From graph, New resistance of thermistor = 2.0 kΩ Since the power delivered is the same, Total resistance before =total resistance after $\Rightarrow$ 3100+1200 = R +2000 $\Rightarrow$ R = 2 300 Ω	**B1** **M1** **A1**
	[Total : 8 marks]	